Pauli Operations

Properties

n-qubit Pauli Operation

→ Pauli Operations as Generators

→ say $P_1, \ldots, P_r$ are n-qubit Pauli Operations

$\langle P_1, \ldots, P_r \rangle$ → set generated by all matrices we obtain by multiplying $P_1, \ldots, P_r$ in any order, any number of times

Ex. $\to \langle X, Y, Z \rangle = \{ \alpha P : \alpha \in \{1, -i, -1, -i\}, P \in \{I, X, Y, Z\}\}$

Pauli Observables

→ Hermitian Matrix Observables

$$A = \sum_{k=1}^{m} \lambda_k P_k \qquad \sum_{k=1}^{m} P_k = I$$

$\lambda_k \to$ Eigenvalue

$P_k \to$ Projections

$A \to$ Projective measurement described by λ_k, P_k

$A = A^\dagger \to A$ is Hermitian Matrix

→ Measurement from Pauli Operations

$X = |+\rangle\langle+| - |-\rangle\langle-|$

$Y =$

$Z =$

Eigenvalues → ± 1 → measurement outcomes

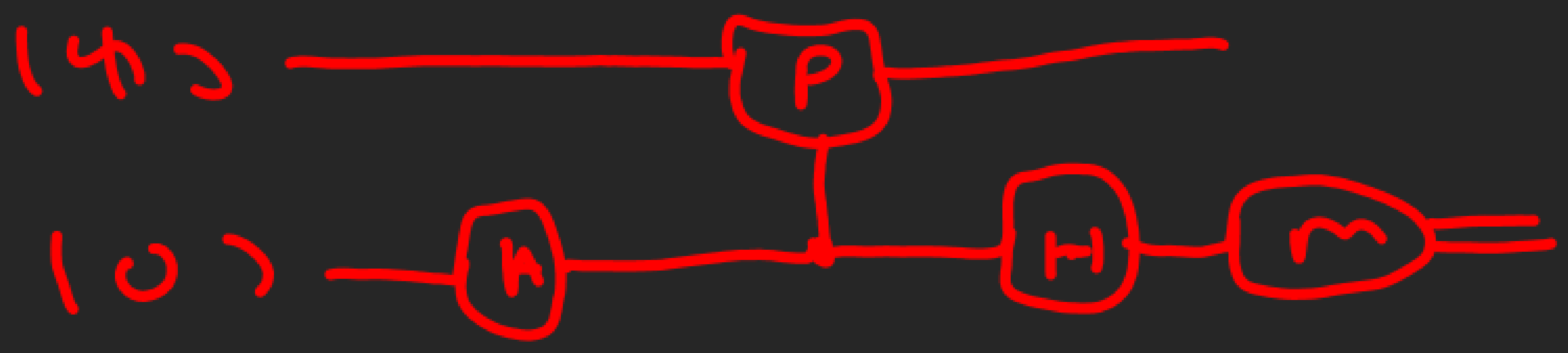
Eigenvectors → $\{|\pm\rangle\}$, $\{|\pm i\rangle\}$, $\{|0\rangle, |1\rangle\}$ → Projection

→ For any non-Identity n-qubit Pauli operation, the 2^n dimensional state space always splits into 2 subspaces of eigenvectors having equal dimensions

→ Both of these projections will have rank 2^{n-1}

$\text{ran}(\rho)$ → No. of non-zero eigenvalues of ρ

→ Non-Destructive Implementation through Phase Estimation



$$P = |\psi_0\rangle\langle\psi_0| - |\psi_1\rangle\langle\psi_1|$$

$$P|\psi_0\rangle = |\psi_0\rangle$$
$$P|\psi_1\rangle = -|\psi_1\rangle$$

$$|\psi\rangle = \alpha|\psi_0\rangle + \beta|\psi_1\rangle$$

$$|\psi\rangle|0\rangle \rightarrow |\psi\rangle|+\rangle \rightarrow \frac{|\psi\rangle|0\rangle + P|\psi\rangle|1\rangle}{\sqrt{2}}$$

$$= \frac{(\alpha|\psi_0\rangle + \beta|\psi_1\rangle)|0\rangle + (\alpha|\psi_0\rangle - \beta|\psi_1\rangle)|1\rangle}{\sqrt{2}}$$

Applying H,

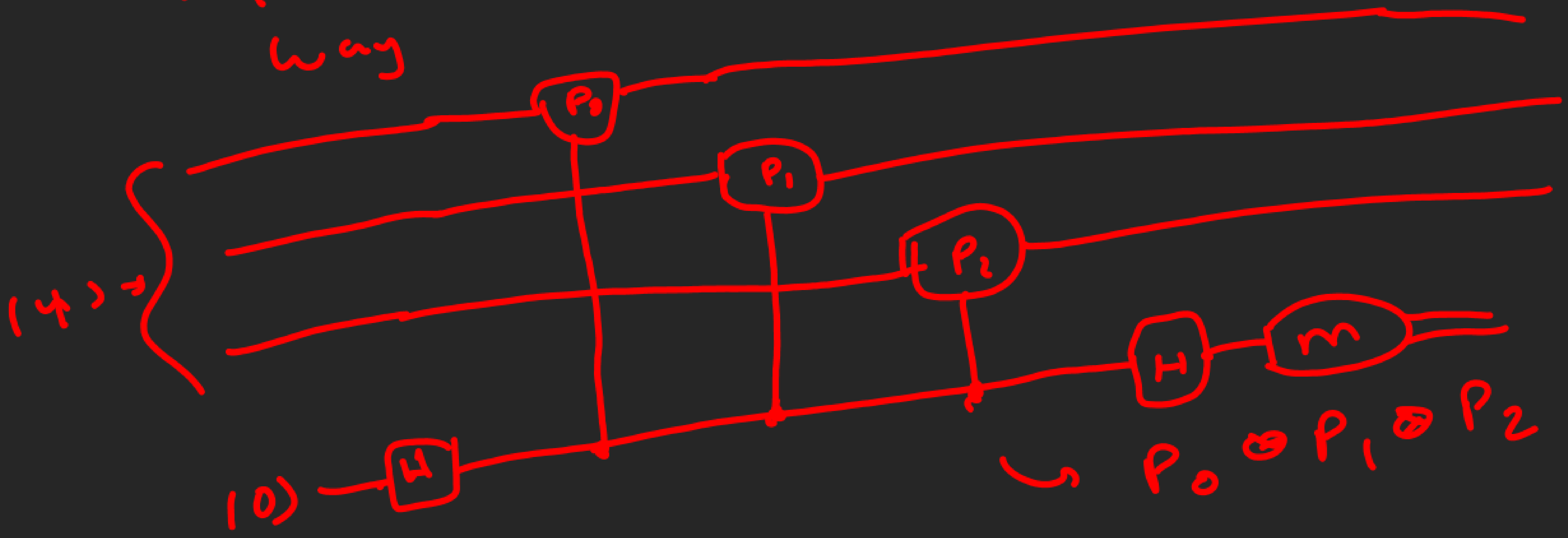
$\alpha|\psi_0\rangle|0\rangle + \beta|\psi_1\rangle|1\rangle$

$p(0) = |\alpha|^2 \rightarrow +1$ EV

$p(1) = |\beta|^2 \rightarrow -1$ EV

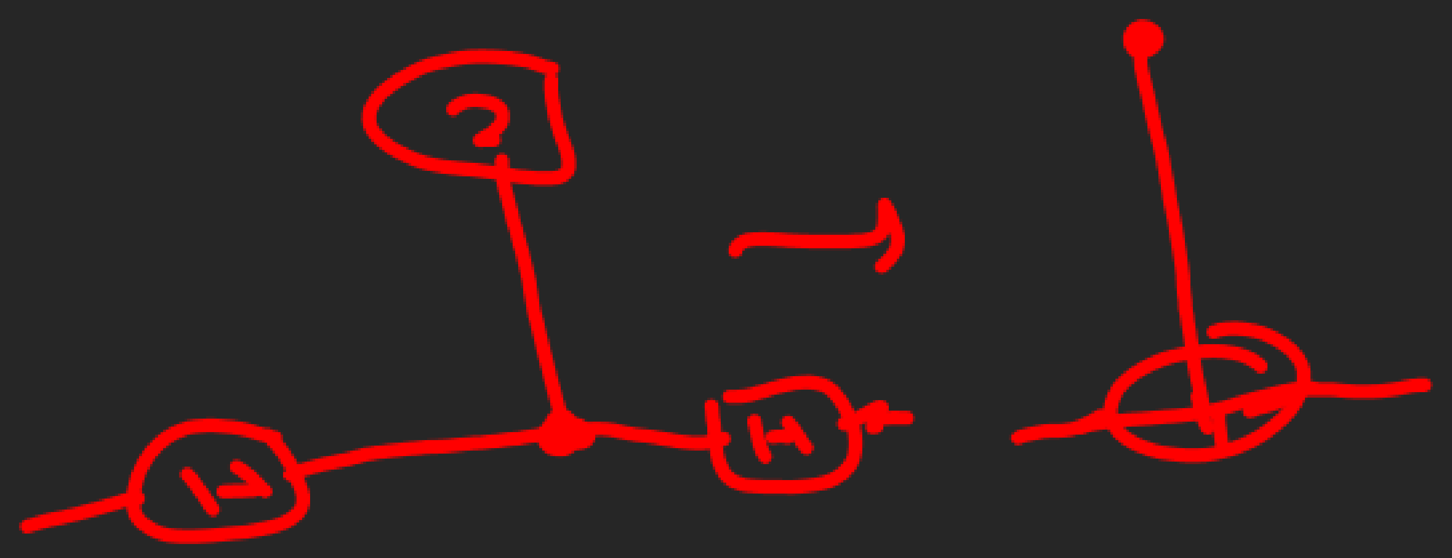
Posterior State $\rightarrow |\psi_0\rangle$ OR $|\psi_1\rangle$

→ For n-qubit Pauli operation, $\lambda = \pm 1$, so it works in the same way



$P_0, P_1, P_2 \in \{X, Y, Z\}$

For, $P_0, \_\_\_ P_r = 2$



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